A continuous cubic obstruction to the C*-algebraic Smale conjecture

Research proof and Lean formalization

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Abstract

We construct a split cubic in $C([0,1],\mathbb{C})$ for which no critical element satisfies the norm form of the C*-algebraic Smale mean-value conjecture. The derivative also splits, both critical elements exist, and the derivative at the test element is a nonzero invertible constant. The obstruction is that connectedness prevents the choice of critical branch from changing between the two endpoints.

Krishna's Conjecture 2.1 in [1] asserts that, for a split polynomial P of degree $n \geq 2$ over a commutative C*-algebra and $P'(z) \neq 0$, some critical element b satisfies

$$\frac{\|P(z) - P(b)\|}{\|z - b\|} \leq \left\|P'(z)\right\|, \tag{1}$$

or the sharper bound with coefficient $(n-1)/n$ on the right. Here P' is the sum of products obtained by deleting one factor. We refute already (1). The source's degree-two theorem and the classical scalar Smale conjecture are unaffected.

Theorem 1. *Let $\mathcal{A} = C([0,1],\mathbb{C})$ with its supremum norm, and put*

$$y(t) = 2t - 1, \qquad q(t) = \sqrt{y(t)^2 + \tfrac{1}{16}}, \qquad z = y.$$

For the monic split cubic

$$P(w) = w(w - \sqrt{3}\,q)(w + \sqrt{3}\,q) = w^3 - 3q^2 w, \tag{2}$$

the critical elements are exactly q and $-q$. The derivative at z is the constant $-3/16$, and neither critical element satisfies (1).

Proof. Expansion of the literal deletion derivative gives

$$P'(w) = 3(w - q)(w + q), \qquad P'(y) = -\tfrac{3}{16}\,1. \tag{3}$$

Consequently both q and $-q$ are critical. Conversely, if $P'(b) = 0$, evaluation at each t in the scalar field gives $b(t) = q(t)$ or $b(t) = -q(t)$. Since $q(t) > 0$, the continuous function b/q takes values in $\{-1, 1\}$. Connectedness of $[0,1]$ makes it constant. Thus $b = q$ or $b = -q$ globally. This argument uses scalar evaluation before factor cancellation; it does not treat $C([0,1],\mathbb{C})$ as an integral domain.

We have $\|y\| \leq 1$ and $\|q\| \leq 5/4$, so for either critical element

$$\|y - b\| \leq \tfrac{9}{4}, \qquad \left\|P'(y)\right\| \|y - b\| \leq \tfrac{27}{64}. \tag{4}$$

Write $r = \sqrt{17}/4$. At $t = 0$, $y = -1$, $q = r$, and

$$P(y)(0) - P(q)(0) = \tfrac{35}{16} + 2r^3 > 2.$$

At $t = 1$, $y = 1$, $q = r$, and

$$P(y)(1) - P(-q)(1) = -\big(\tfrac{35}{16} + 2r^3\big).$$

Hence $\|P(y) - P(b)\| > 2$ for every critical element, contradicting (4) if (1) were true. The denominator is positive: $y = b$ would imply $P'(y) = 0$, contrary to (3). Finally the inverse of $P'(y)$ is the constant $-16/3$. □

Scope and formal certificate. The counterexample lies in a connected commutative unital C*-algebra. All three roots in (2) and both distinct critical elements are continuous. Thus it does not rely on nonexistence of critical elements or a zero derivative at some test fiber. At individual fibers a suitable scalar critical point can change sign; a global continuous critical element cannot.

The Lean certificate contains 46 declarations in five modules, including four reused general continuous-selection lemmas. It uses Lean 4.34.1 and mathlib revision

`d13f23b723b8a846827a245b89c10fc7d3f11612`.

The literal product, deletion derivative, monicity, exact degree, supremum-norm bounds, branch completeness and explicit inverse are proved. The closed endpoint

`CStarSmale.source_counterexample`

negates the universal norm assertion over commutative C*-algebras and all $n \geq 2$. The isolated strict replay passes with only `propext`, `Classical.choice` and `Quot.sound`.

Independent source, prior-art and substantive-eligibility review is pending; no discovery credit or worldwide priority is claimed. The source's separate higher-order and dual mean-value conjectures, known public refutations of those statements, the general selection principle and helper declarations receive no credit here. This manuscript addresses only the norm Smale assertion (1); an order-strong extension is not a separate outcome.

References

[1] K. Mahesh Krishna, *C*-algebraic Smale Mean Value Conjecture and Dubinin–Sugawa Dual Mean Value Conjecture*, arXiv:2206.08154v1 (2022), Conjecture 2.1, pp. 4–5. arXiv:2206.08154v1.